AP PRECALCULUS

Unit 1 • Lesson 1.9-1.10 • Guided notes

Topic 1.9-1.10 • Rational Functions and Asymptotes

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Locate vertical asymptotes and holes of a rational function from its factored form.
- Describe behavior near a vertical asymptote with one-sided limits, determining signs.
- Find the coordinates of a hole.

Vocabulary. vertical asymptote, hole, one-sided limit, $\lim_{x\to a^-}$, $\lim_{x\to a^+}$, multiplicity, simplified form.

1. Vertical asymptotes or holes?

Key idea

Factor numerator and denominator completely. For a zero a of the denominator:

- if $(x-a)$ appears _____ times in the denominator than in the numerator, $x=a$ is a _____;
- if it appears _____ times in the numerator, there is a _____ at $x=a$.

2. Behavior near a vertical asymptote

Near $x=a$, $|r(x)|$ grows _____. To decide $+\infty$ or $-\infty$, check the _____ of each factor just left and just right of a. If $(x-a)$ remains an _____ number of times in the denominator, the sign flips across the asymptote; an _____ number of times, it stays the same.

Example 1

Describe $r(x)=\dfrac{x+1}{(x-2)^2}$ and $s(x)=\dfrac{x-1}{(x+2)(x-3)}$ near their vertical asymptotes.

3. Holes

Key idea

To find the hole at $x=a$: cancel the common factors, then substitute a into the _____ expression. The hole is at $(a,\ \text{that value})$, and $\lim_{x\to a} r(x)$ equals _____.

Example 2 • Full analysis

Analyze $r(x) = \dfrac{x^2 - x - 6}{x^2 - 9}$: hole, vertical asymptote, zero, end behavior. Then sketch.

Example 3 • A hole on the axis

$r(x) = \dfrac{(x-1)^2}{x-1}$. Is $x = 1$ a zero?

AP exam tip

Justify each feature with the factored form *and* a limit: "$\lim_{x \to -3^+} r(x) = -\infty$, so $x = -3$ is a vertical asymptote"; "$\lim_{x \to 3} r(x) = \frac{5}{6}$ and $r(3)$ is undefined, so there is a hole at $\left(3, \frac{5}{6}\right)$."

Common mistakes

- Calling every zero of the denominator a vertical asymptote without checking for cancellation.
- Finding the hole's height in the original expression ($\frac{0}{0}$).

Check your understanding

1. $\dfrac{x-3}{x^2-9}$ has a vertical asymptote at _____ and a hole at _____.
2. $\lim_{x \to 0^-} \dfrac{2}{x^3} =$ _____.